

The Grand Aurelia Hotel – Reservation System

A command-line hotel room reservation and billing system written in Python.

Features

- View all available rooms in an aligned table
- Select a room by **room number** or **room type** (Single, Double, Deluxe, Suite)
- View full room details (bed, balcony, view, facilities)
- Multi-guest support (1 up to the room's maximum capacity)
- Optional add-on services: Breakfast, Food, Laundry, Room Service
- Special request field
- Automatic bill with 18% GST
- Payment method selection: Cash, UPI, Card
- Final confirmation, with room released if the booking is cancelled
- Input validation (numbers, guest limits, menu choices)

Requirements

- Python 3.6 or above (uses f-strings)
- No external libraries

How to Run

```
python hotel.py
```

(Replace `hotel.py` with whatever you named the file)

Program Flow

1. Show available rooms
2. Select room → view details → confirm selection
3. Enter guest details and number of nights
4. Choose optional services
5. Enter special request (optional)
6. View final bill
7. Choose payment method
8. Confirm or cancel booking

Room Data

Room	Type	Price/Night	Max Guests	Bed	View
1 01	Single	Rs. 1 200	1	Single Bed	Garden View
1 02	Single	Rs. 1 200	1	Single Bed	Garden View
1 03	Double	Rs. 1 800	2	Queen Bed	City View

Room	Type	Price/Night	Max Guests	Bed	View
1 04	Double	Rs. 1 800	2	Queen Bed	City View
1 05	Deluxe	Rs. 2500	2	King Bed	Garden View
1 06	Suite	Rs. 3500	3	King Bed	Premium View

Services

Service	Price
Breakfast	Rs. 250
Food	Rs. 300
Laundry	Rs. 200
Room Service	Rs. 1 50

Billing Formula

Room Charge = Price per night × Nights

Subtotal = Room Charge +

Service Charges GST = Subtotal

Sample Output

THE GRAND AURELIA HOTEL RESERVATION SYSTEM

```
Room      : 103
Nights    : 2
Room Charge : Rs.
3600 Services :
Rs. 250 Subtotal :
Rs. 3850
GST (18%) : Rs.
693.0 TOTAL :
Rs. 4543.0
```

Concepts Used

Dictionaries, lists, functions, **while** loops, loops, **if/elif/else**, **try/except**, string formatting (f-strings), input validation, recursion.

Limitations

- Handles one booking per run
- Data is not saved to a file or database
- No check-in/check-out dates
- Phone number is not validated

Future Improvements

- Save bookings to a file / database
- Date-based availability

- Multiple bookings and admin menu
- GUI or web version

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